

M4 — Exploratory Data Analytics Report

Project: LZU INFO422 — Influencing Factors behind Video Game Sales

Group 23 | Week 7

Author: Zhong Rui

Dataset: `c1eanned_vgchartz.csv` (8,786 records × 33 features)

1. Executive Summary

EDA of the VGChartz 2024 dataset investigates three research questions via 12 visualizations.

Key findings:

- Sales are extremely right-skewed (skewness = 8.04); `log_sales` reduces this to 2.55.
- Platform-genre interactions matter: Shooter×X360 and RPG×PS2 outperform.
- Regional preferences are distinct: Japan favors RPG, NA favors Shooter/Sports, PAL favors Racing.
- Critic score has weak correlation with sales (Spearman $\rho \approx 0.05$).
- K-Means (K=4) isolates a blockbuster tier (2.7% of games, median 3.58M).
- Publisher brand effects operate through **scale** (volume) and **prestige** (per-game premium).

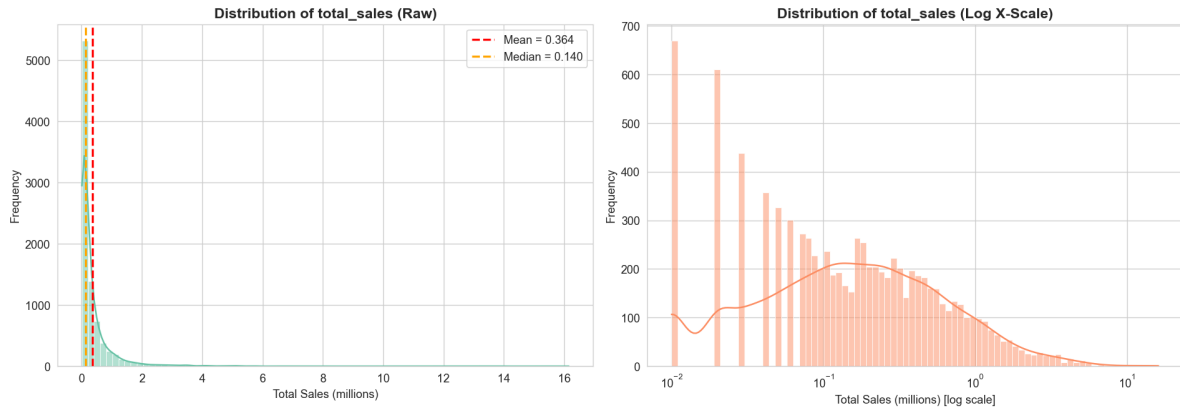
2. Data Overview

Category	Columns	Count
Categorical	console (33), genre (20), publisher (551), developer	4
Core numerical	total_sales, critic_score, 4 regional sales	6
Temporal	release_date, release_year, release_year_bin (5 eras)	3
Clustering	sales_cluster (K=4)	1
Derived target	log_sales = $\ln(1 + \text{total_sales})$	1
Brand proxies	publisher_game_count, developer_game_count	2

Data quality: Core features have 0% missing values. `jp_sales` has 63.7% zeros (post-imputation), flagged via `has_regional_data`.

3. Univariate Analysis

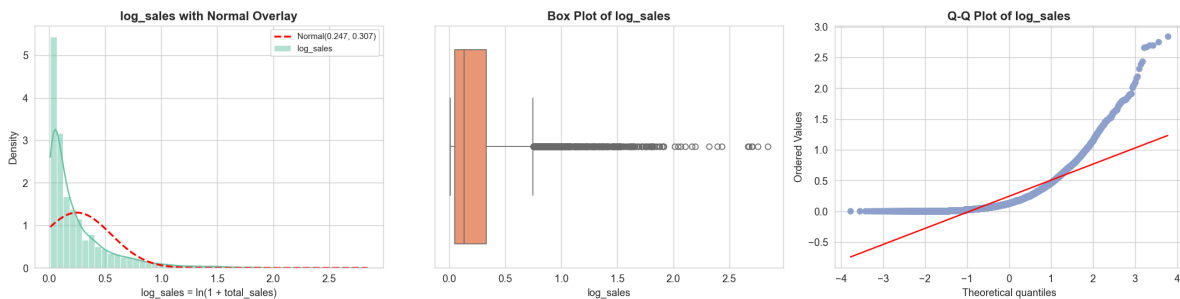
Visualization 1: Raw Sales Distribution



Viz 1 — Raw `total_sales`: Mean (0.364M) is 2.6× the median (0.140M); skewness = 8.04. Over 75% of games sell below 0.39M; maximum = 16.15M. The log-scale view confirms sales span multiple orders of magnitude.

Takeaway: `log_sales` should be the modelling target to avoid outlier domination.

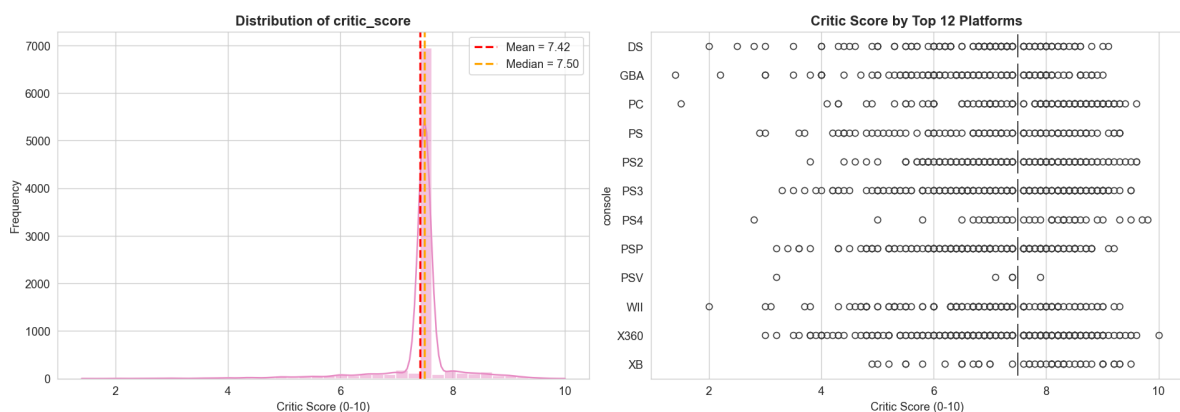
Visualization 2: `log_sales` Distribution & Normality Check



Viz 2 — `log_sales`: Skewness drops 68% (8.04 → 2.55). Q-Q plot shows good linearity in the central 95%. Shapiro-Wilk rejects exact normality, but the distribution is well-behaved for parametric methods.

Takeaway: `log_sales` is a viable regression target.

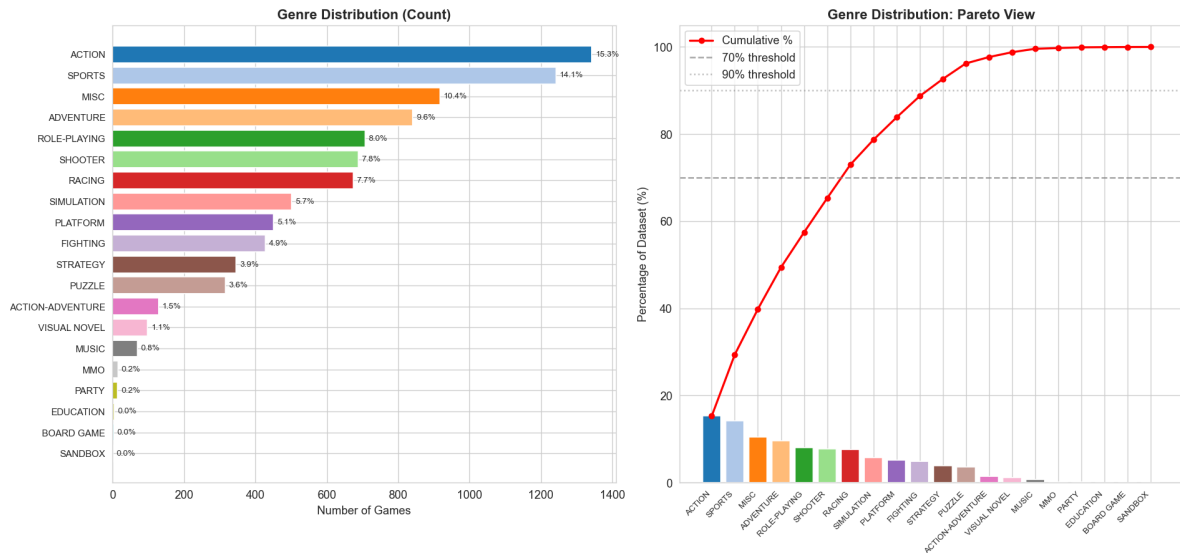
Visualization 3: Critic Score Distribution



Viz 3 — `critic_score`: Mean = 7.42, median = 7.50. A spike at 7.50 reflects median imputation of ~2,300 missing scores. Platform medians all cluster near 7.5, suggesting review aggregation normalizes scores across the industry.

Takeaway: `critic_score` has limited variance and an imputation artifact; not a strong standalone predictor.

Visualization 4: Genre Class Balance

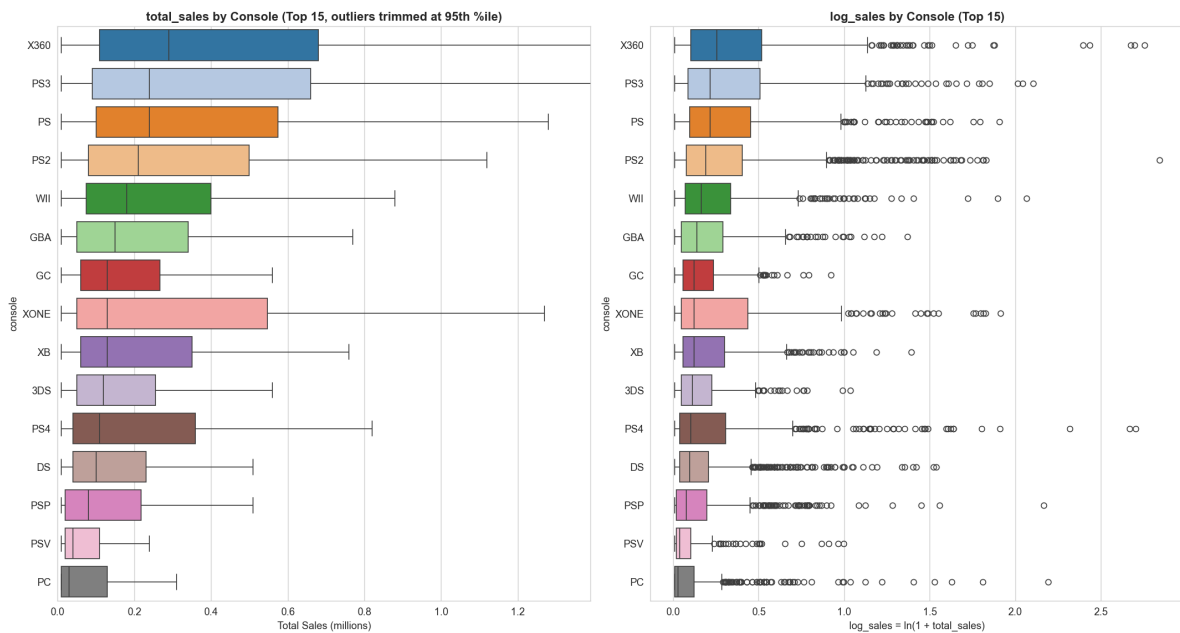


Viz 4 — Genre balance: Top 3 genres (ACTION 15.3%, SPORTS 14.1%, MISC 10.4%) = 39.8%. Top 7 = 70.7%. Tail includes SANDBOX (1), BOARD GAME (2), EDUCATION (2).

Takeaway: Rare genres (<1%) should be aggregated as "Other" in per-genre analyses.

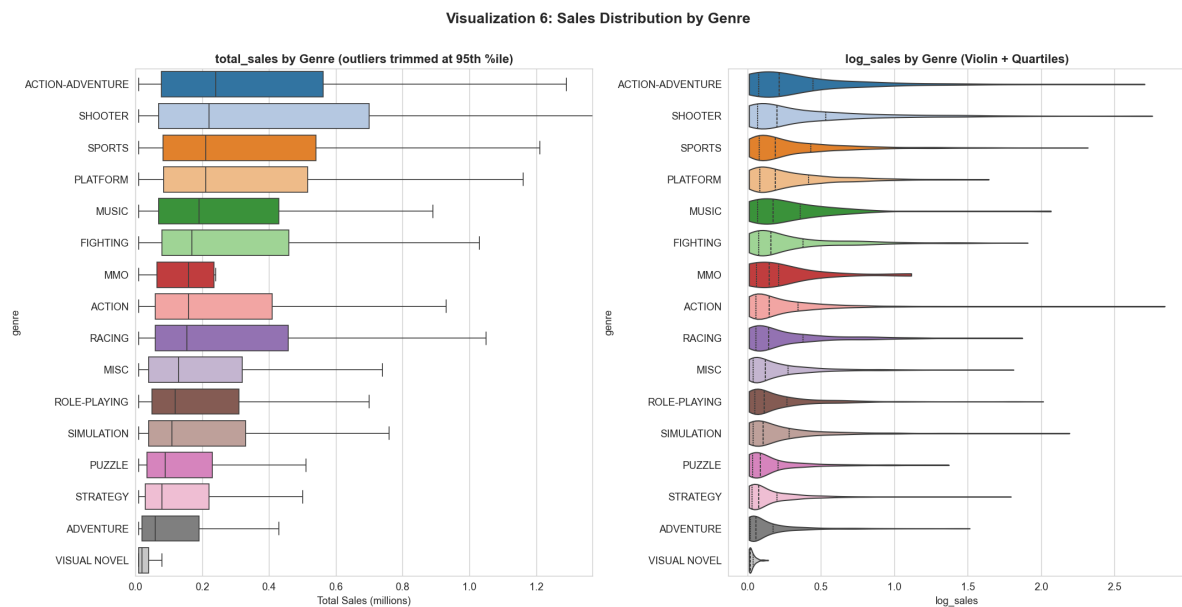
4. Bivariate Analysis

Visualization 5: Sales Distribution by Console Platform



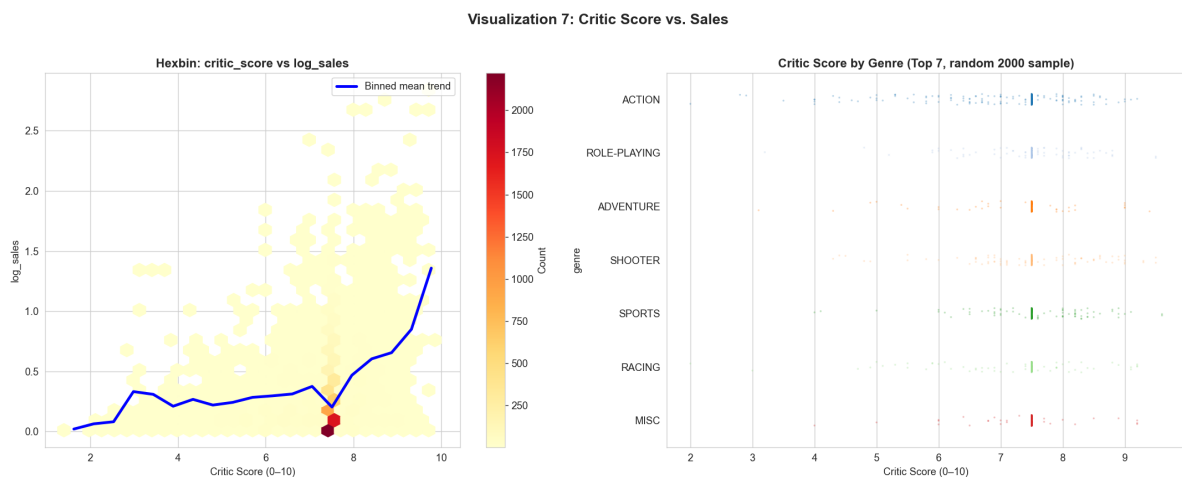
Viz 5 — Sales by platform: DS and PS2 lead in game count (~1,055 each). Median sales are uniform (0.05M–0.20M). Upper-tail differences matter: PS4, X360, and Wii show elevated outliers. PC has the lowest median, likely from undercounted digital distribution.

Takeaway: Platform alone weakly differentiates median sales; the platform-genre interaction (Viz 8) is more informative.



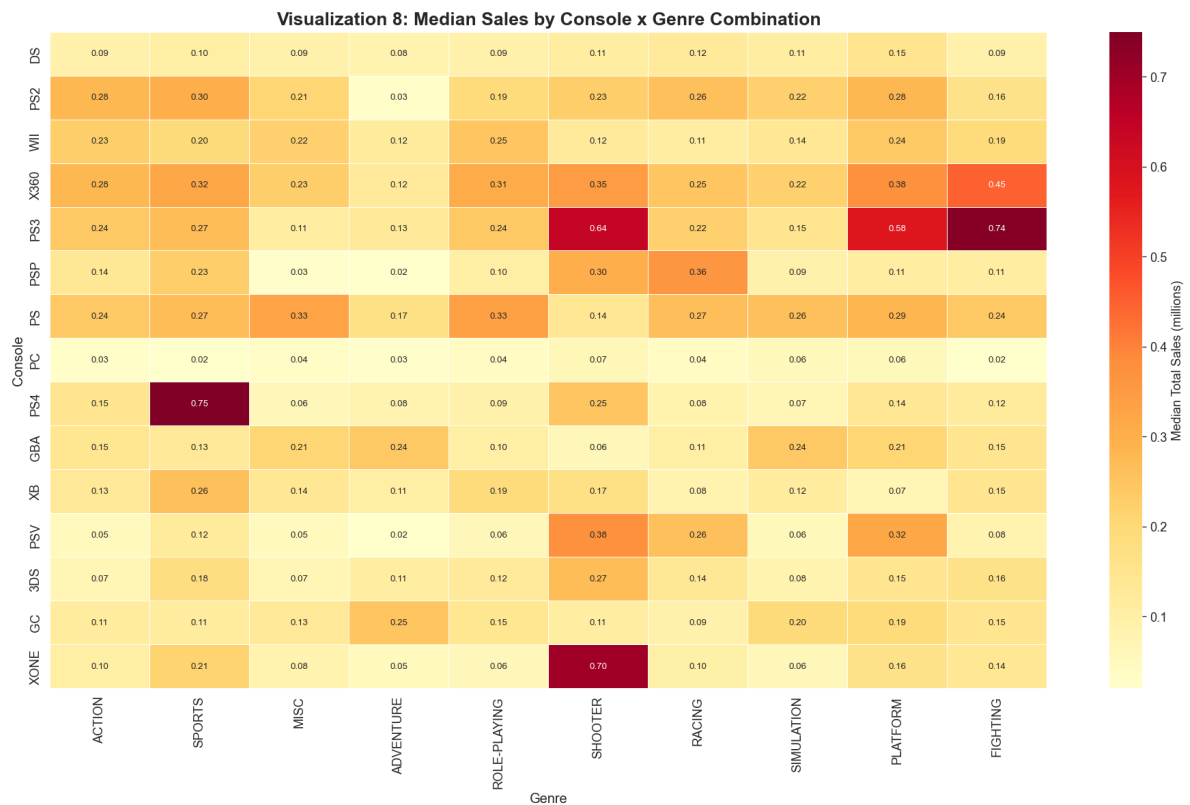
Viz 6 — Sales by genre: SHOOTER and ROLE-PLAYING have the highest medians and widest spread (blockbuster economics). SPORTS is compressed (annualized franchises). MISC is bimodal — a catch-all mixing niche and hits.

Takeaway: Genre differentiates sales more strongly than platform. SHOOTER/RPG carry a premium; niche genres have a low ceiling.



Viz 7 — Critic score vs. sales: Weak positive correlation (Pearson $r \approx 0.05$, Spearman $\rho \approx 0.05$). Sales spread at any given score is enormous — even perfect 10.0 scores span the full range.

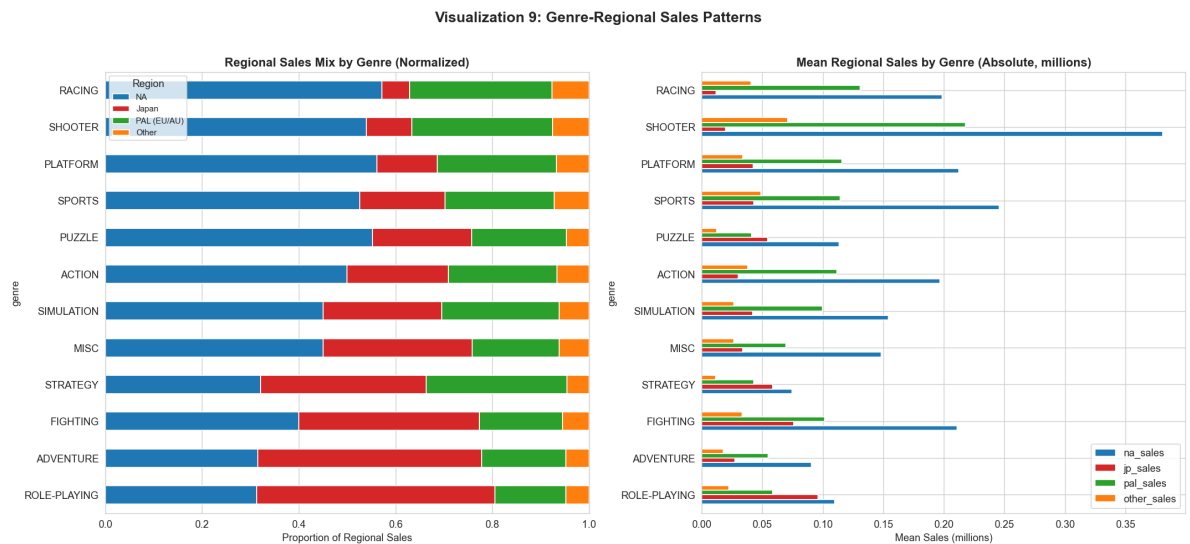
Takeaway: Critic score alone explains $<0.5\%$ of variance. Not a primary predictor.



Viz 8 — Console × genre heatmap: Top combinations: PS4×SPORTS (0.75M), PS3×FIGHTING (0.74M), PS3×SHOOTER (0.64M), XONE×SHOOTER (0.70M). PC is consistently low (0.02–0.07M).

Takeaway: Platform-genre interaction is meaningful. Include interaction terms for Q1 modelling.

5. Multivariate Analysis



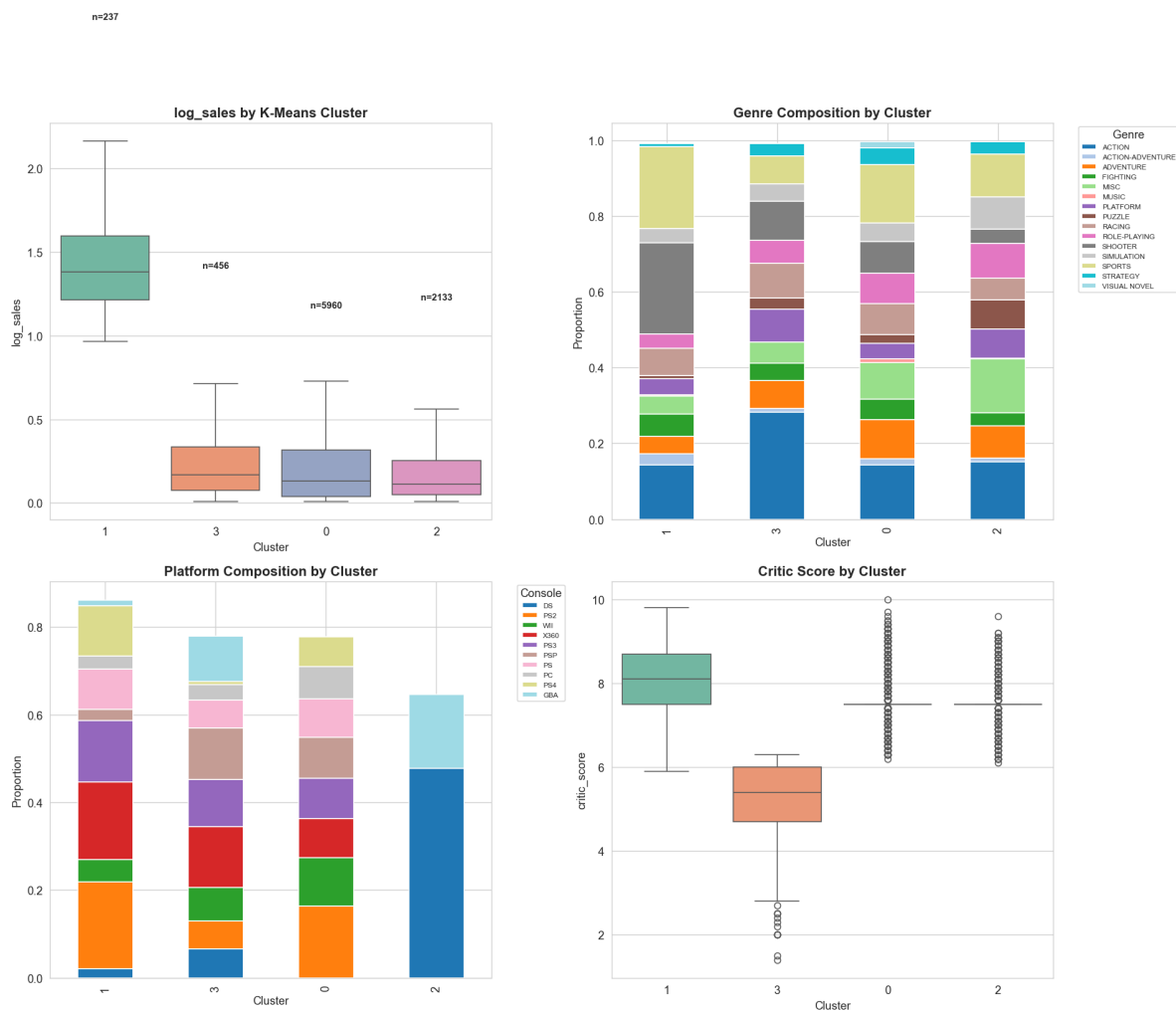
Viz 9 — Regional patterns: Three distinct clusters:

Region	Dominant Genres	Key Pattern
NA	SHOOTER, SPORTS, ACTION	na_ratio > 0.40
Japan	RPG, VISUAL NOVEL, FIGHTING	jp_ratio > 0.35 for RPG
PAL	RACING, PLATFORM, SPORTS	Elevated pal_ratio

NA dominates absolute volume, but proportional analysis reveals genuine cultural preferences.

Takeaway: Regional ratios are viable inputs for region-specific models (Q2).

Visualization 10: K-Means Cluster Characterization



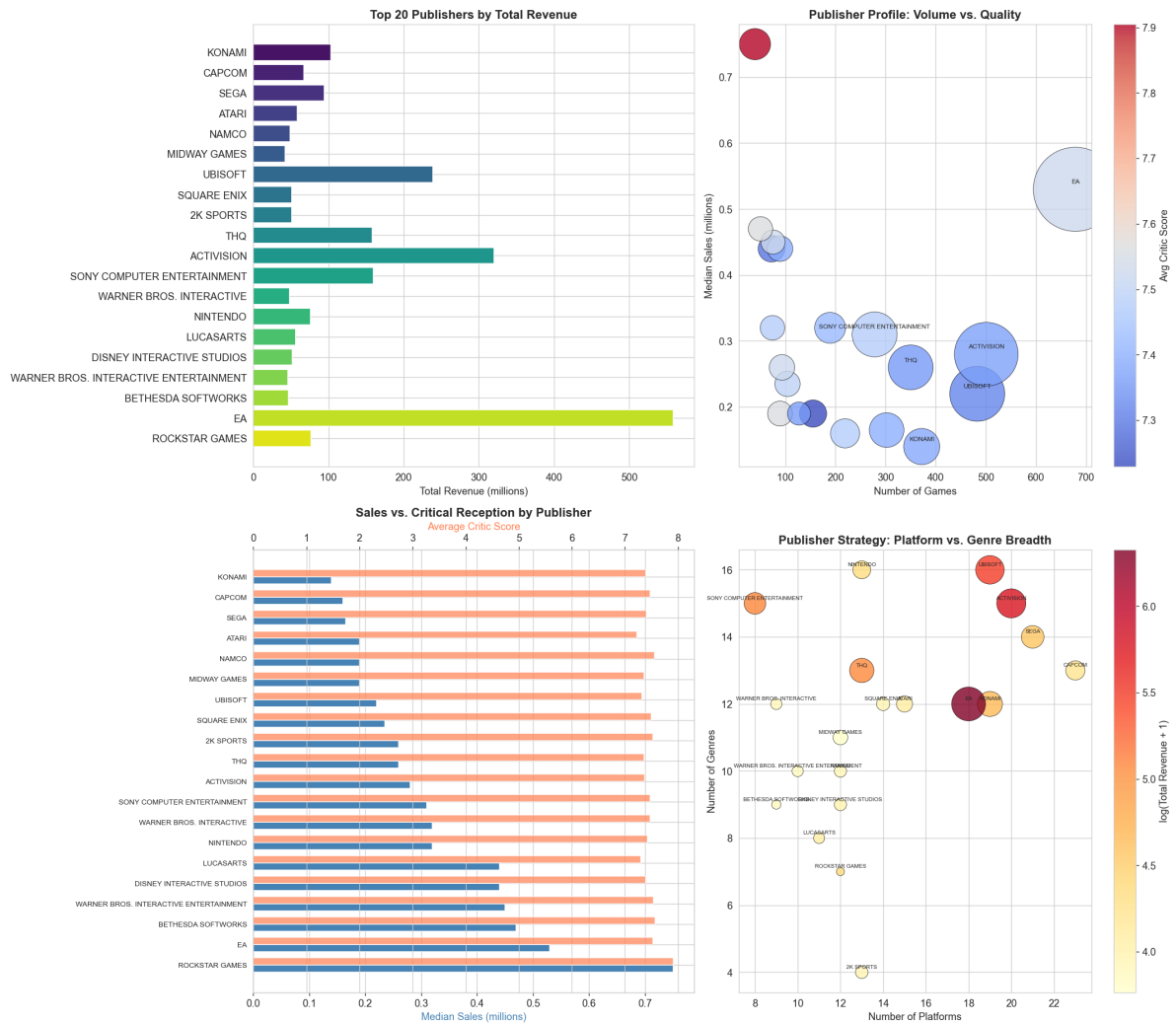
Viz 10 — K-Means clusters (K=4):

Cluster	n	%	Median	Dominant Genres	Tier
1	237	2.7%	3.58M	ACTION, SHOOTER	Blockbuster
0	5,960	67.8%	0.29M	Balanced	Mainstream
3	456	5.2%	0.33M	SPORTS, RACING	Mid-tier
2	2,133	24.3%	0.24M	MISC, ADVENTURE	Budget/Niche

Critic scores vary minimally across clusters (7.5–8.0).

Takeaway: Enables a two-stage approach (classify tier → predict within-tier), addressing target imbalance.

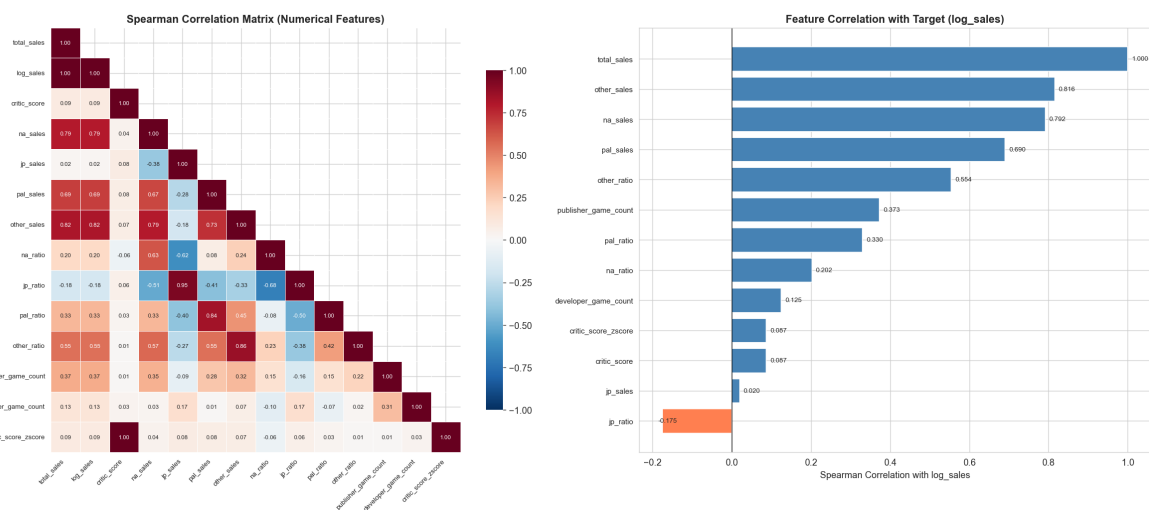
Visualization 11: Publisher Brand Analysis (Top 20)



Viz 11 — Publisher brand: Volume publishers (EA, Ubisoft: many games, moderate per-title sales) vs. **prestige publishers** (Nintendo, Rockstar: fewer games, higher premium). Nintendo achieves both. Critical and commercial success are loosely coupled.

Takeaway: Brand effect operates through scale (`publisher_game_count`) and prestige (publisher identity). Include both separately for Q3.

Visualization 12: Correlation Structure & Feature-Target Relationships



Viz 12 — Correlation matrix: Western regional sales are highly intercorrelated ($p > 0.6$). `jp_sales` is weaker ($p \approx 0.3$ – 0.4), confirming Japan's distinctiveness. `publisher_game_count` and `developer_game_count` are highly correlated ($p > 0.8$) — only one should enter a model.

Takeaway: Avoid data leakage (regional sales alongside `total_sales`). Japan-Western divergence justifies region-specific sub-models (Q2).

6. Revised Hypotheses

Q1 — Platform-Genre Sales Advantage

Original: Total sales vary across platform-genre combinations.
Revised (supported): The interaction is meaningful (Viz 5, 6, 8), concentrated in the upper quartile. SHOOTER/RPG carry a premium. Handhelds excel in niche genres. Later-gen consoles show higher per-game sales.

Q2 — Genre-Regional Impact

Original: Genres impact sales differently across regions.
Revised (strongly supported): Three clusters identified (Viz 9): NA→Shooter/Sports/Action; JP→RPG/Visual Novel/Fighting; PAL→Racing/Platform/Sports. Limitation: `jp_sales` has 63.7% zeros.

Q3 — Brand Effect

Original: Brand significantly impacts sales.
Revised (supported): Two mechanisms: (1) **scale** via `publisher_game_count`, (2) **prestige** via publisher identity. Model separately.

7. Modelling Question

"Can we predict a video game's global sales (`log_sales`) using platform, genre, publisher, critic score, and release era, and quantify each factor's relative importance?"

Component	Specification
Target	<code>log_sales</code>
Features	console, genre (one-hot), publisher, developer (target-encoded), critic_score, release_year_bin, sales_cluster
Interactions	Platform-genre term, regional ratios
Brand proxies	<code>publisher_game_count</code> OR <code>developer_game_count</code>
Models	Linear Regression → Random Forest → XGBoost
Evaluation	R ² , RMSE, MAE; 5-fold CV
Validation	Feature importance ranking for Q1–Q3

Q1: Random Forest feature importance. Q2: Regional ratios + stratified evaluation. Q3: Compare publisher identity vs. scale.

Appendix: Visualization Inventory

#	Type	Visualization	Addresses
1	Univariate	Sales distribution (raw + log)	—
2	Univariate	<code>log_sales</code> + QQ-Plot	—
3	Univariate	<code>critic_score</code> by platform	—
4	Univariate	Genre balance (Pareto)	—
5	Bivariate	Sales by console	Q1
6	Bivariate	Sales by genre	Q1, Q2
7	Bivariate	Critic score vs. sales	—
8	Bivariate	Console × genre heatmap	Q1
9	Multivariate	Regional proportions	Q2
10	Multivariate	K-Means clusters	Q1, Q3
11	Multivariate	Publisher brand	Q3
12	Multivariate	Correlation matrix	M5

Total: 12 visualizations

References

- VGChartz 2024 Dataset: [Kaggle](#)
- Full EDA code: `INFO422-DS Project-group-23-EDA.ipynb`